

Data Flow Diagram

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Team ID	SWTID-2026-4836
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram

the DFD illustrates how farmers provide soil and environmental information, how the machine learning system processes the data, and how the recommended crop is generated and displayed to the user.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A user outside the system who sends or receives data, such as the Farmer who enters soil and environmental details and receives crop recommendations.
Rectangle with numbered header	Process	An activity that processes agricultural data, such as Enter Soil Data, Crop Recommendation Processing, and Display Prediction Result.
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, such as soil parameters, prediction results, and crop recommendations.

Example

The following DFD represents the flow of data in the OptiCrop system. It shows how soil and environmental parameters entered by the farmer are processed by the machine learning model using the agricultural dataset to generate crop recommendations and store prediction records.

